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One-hour post-load plasma glucose level during the OGTT predicts dysglycemia Observations from the 24 year follow-up of the Israel Study of Glucose Intolerance, Obesity and Hypertension

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ABSTRACT

Aims: The present study assessed the longitudinal association of an elevated 1-h plasma glucose [1-h-PG >8.6 mmol/l (155 mg/dl)] with and without impaired glucose tolerance [IGT; 2-h-PG 7.8–11.0 mmol/l (140–199 mg/dl)] with cumulative incident of diabetes and prediabetes over 24 years in a non-diabetic cohort.
Methods: From 1979 to 1984, 1970 non-diabetic men and women completed an oral glucose tolerance test (OGTT), physical and biochemical measurements as well as a questionnaire related to lifestyle and medical background. During the years 2000–2004, 853 survivors of the original cohort were interviewed and re-examined for glycemic progression.
Results: Individuals with 1-h-PG >8.6 mmol/l (155 mg/dl) but with 2-h-PG <7.8 mmol/l (140 mg/dl) had a significantly elevated risk, compared to those with both 1-h-PG ⩽8.6 mmol/l (155 mg/dl) and 2-h-PG <7.8 mmol/l (140 mg/dl), for both diabetes [OR:4.35 (95%CI: 2.50–7.73)] and prediabetes outcomes [OR:1.87 (95%CI 1.09–3.26)], adjusted for sex and age, smoking, body mass index, blood pressure, fasting blood glucose and insulin.
Conclusions: The risk for diabetes associated with a 1-h level >8.6 mmol/l (155 mg/dl) is increased and further worsened in the presence of IGT. Identifying individuals at risk with a 1-h-PG glucose level during an OGTT is recommended.
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1. Introduction

Early detection of individuals at high-risk of developing type 2 diabetes (T2DM) is of paramount importance for the prevention of T2DM *per se* and cardiovascular disease. Prediabetes, impaired fasting glucose [IFG; ⩾5.6–7.00 mmol/l (100–125 mg/dl)] and impaired glucose tolerance [IGT; 2 h PG ⩾7.8–11.0 mmol/l (140–199 mg/dl)] and HbA1c [⩾5.7–6.4% (39–46 mmol/mol)], represents an intermediate stage in the transition from normal glucose tolerance (NGT) to overt T2DM [1,2]. Prospective epidemiological studies have demonstrated that ~40% of those who develop T2DM had NGT at baseline [3]. Furthermore, findings from several observational studies [3] have shown that prediabetic conditions as defined by fasting plasma glucose (FPG), 2 h plasma glucose derived from oral glucose tolerance test and HbA1c are limited in their ability to identify individuals at future risk for developing diabetes [4,5].

Plasma glucose levels measured at earlier time points (i.e., 30 min and 1-h) during the oral glucose tolerance test (OGTT) have been shown to identify individuals at increased risk for T2DM [5,6]. Large scale epidemiological studies demonstrated that 1-h PG >155 mg/dl during the OGTT were associated with increased insulin resistance, decreased β-cell function, T2DM, CVD, kidney disease and mortality [7–10]. Using a 100 g OGTT, we have previously demonstrated an association of the 1-h PG with long-term mortality over 23 years [7,11]. The present study assessed the longitudinal association of the 1-h PG in the absence and presence of IGT with incident diabetes in a non-diabetic, Israeli population over a mean follow-up period of 24 years.

2. Methods

The Israel Study of Glucose Intolerance, Obesity and Hypertension (The Israel GOH Study) initiated in 1969 is a prospective cohort study randomly drawn from the Israeli Central Population Registry according to age (3 strata of subjects born between 1912–1921, 1922–1931, and 1932–1941), sex (50% from each sex) and ethnic origin (4 main origins: Yemen, North Africa, Middle East, Europe/America, according to country of birth and for an Israeli born – according to country where father was born) [12–14]. This database was established to determine the factors associated with diabetes, cardiovascular risk and mortality within the Israeli population. It also seeks to determine the causes of these conditions in order to provide a rational basis for recommendations towards their prevention. The present report describes the outcome after a 24-year follow-up period.

From 1979 to 1984, referred to as baseline, 1970 individuals without diabetes completed a 2-h, 100 g OGTT and had physical (height, weight, blood pressure), and biochemical (fasting glucose, insulin, and lipid profile) measurements by trained research nurses. Furthermore, they have also completed a questionnaire relating to their lifestyle and medical background. No major differences in age-sex-ethnic origin distributions of the study group compared to the surviving original cohort were noted. A flow chart of survival, inclusion, and follow-up for the current report is illustrated in Fig. 1.

During the years 2000–2004, survivors of the original cohort were interviewed and re-examined for glycemic progression. Of the 1970 individuals who constituted the target population for the current study, 756 were not included: 254 were deceased, 278 were unable to be traced, 177 refused to undergo re-examination and 47 were too ill to participate. Of the remaining 1214 individuals, 177 were excluded from the analysis because they were only interviewed by telephone and did not undergo blood testing to validate their glycemic status. In addition, 184 individuals were classified as having diabetes at baseline and therefore were not included in the present analysis (see flow diagram, Fig. 1).

We compared the 853 participants in the current analysis with 361 non-participants. The latter were 2 years older on

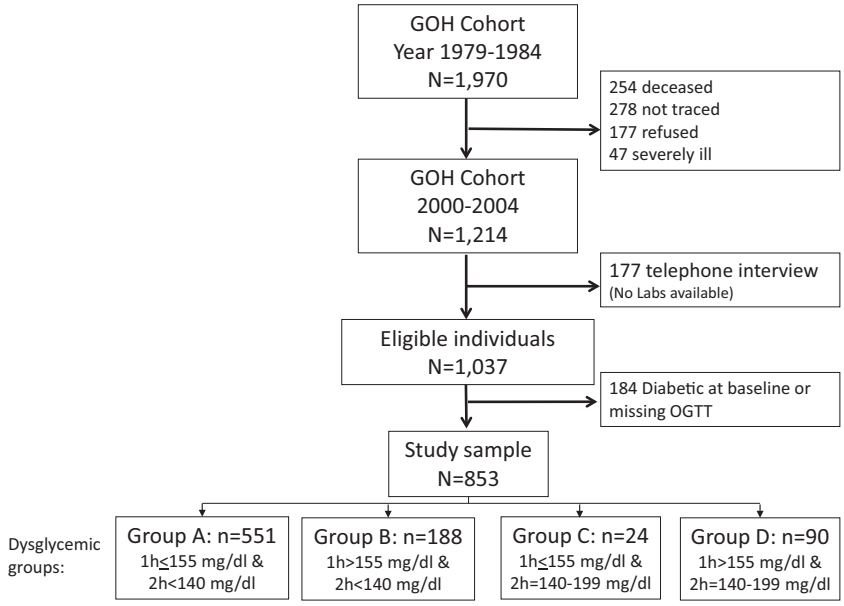

Fig. 1 – Flow diagram of the GOH cohort.

average, predominantly female (63.4% compared to 48.2% in participants), and smoked less (33% and 40% were either previous or current smokers in non-participants and participants, respectively). No differences were observed between non-participants and participants in ethnic origin, blood pressure values, total cholesterol, fasting insulin, HDL-cholesterol, or in triglycerides.

Participants were interviewed by trained nurses in regional clinics and were asked about their lifestyle habits, medical history, and use of medications. Height, weight and three blood pressure measurements were taken during the interview. Venous blood was drawn after a 12 h fast for glucose and lipids (total serum cholesterol, LDL- and HDL-cholesterol, and triglycerides). An OGTT was performed in non-diabetic individuals with 100 g of glucose (instead of the standard 75-gram) as it has been shown to have little effect on glucose levels [15] but enhances insulin responsiveness [16]. Blood glucose was determined after 1 and 2 h.

Plasma glucose was determined by an enzymatic UV test (hexokinase method) performed on a Beckman-Coulter AU5800 Analyzer. Plasma insulin was determined in duplicates by the Phadebas Radioimmunoassay kit (Pharmacia Diagnostics Inc. Piscataway, NJ, CV%: within-assay 4 and the between-assay: 8%) [17]. Plasma glucose and insulin levels were determined fasting and 1- and 2-h after the glucose load. The facilities of Sheba Medical Center were used for laboratory determinations.

The following definitions were applied: *Normoglycemic* — not using glucose lowering medications and FPG <5.6 mmol/l (100 mg/dl), and, when available, 2 h blood glucose <7.8 mmol/l (140 mg/dl); *Prediabetes* – including IFG: FPG 5.6–7.8 mmol/l (100–125 mg/dl); IGT: 2-h blood glucose 7.8–11.0 mmol/l (140–199 mg/dl); *Diabetes (T2D)* — reported diabetes (substantiated by documentation of prescriptions for oral glucose lowering medications or insulin, or medical reports); 2 h blood glucose of ⩾11.1 mmol/l (200 mg/dl) or fasting glucose ⩾7.8 mmol/l (126 mg/dl);

Four categories of baseline post -OGTT glycemic status were established according to 1-h and 2-h glucose levels:

- Group-A = 1 h ⩽155 mg/dl (8.6 mmol/l) and 2 h <140 mg/dl (7.8–11.0 mmol/l);
- Group-B = 1 h >155 mg/dl (8.6 mmol/l) and 2 h <140 mg/dl (7.8–11.0 mmol/l);
- Group-C = 1 h ⩽155 mg/dl (8.6 mmol/l) and 2 h = 140–199 mg/dl (IGT) (7.8–11.0 mmol/l);
- Group-D = 1 h >155 mg/dl (8.6 mmol/l) and 2 h = 140–199 mg/dl (IGT) (7.8–11.0 mmol/l).

Inclusion criteria were: Survival until follow-up with fasting blood glucose <126 mg/dl (7.0 mmol/l) and 1- and 2- h post-load glucose values available at baseline. Individuals with diabetes at baseline were excluded.

The Institutional Review Board of the Sheba Medical Center approved the current study based on the third follow-up of the GOH cohort. All participants provided written consented.

2.1. *Statistical analysis*

Associations in demographic variables (age and sex), smoking and health baseline anthropometric (BMI, blood pressure) and biochemical characteristics (glucose, insulin and plasma lipids) according to the 4 glycemic groups (A,B,C and D) were tested using the Chi-square test for categorical variables and one-way analysis of variance for continuous variables. The Kruskal–Wallis non-parametric test was used for the comparison between the four study groups and the continuous non-normally distributed variables (HOMA-IR and triglyceride levels). Multiple multinomial logistic regression models were fitted to study variables that were significantly and independently associated with diabetes or prediabetes development with normoglycemic category serving as a reference group. Receiver operating characteristic (ROC) curves were plotted to analyze the performance of the 1-h and 2-h plasma glucose on prediabetes and diabetes outcomes. All analyses were two-tailed and performed using SAS (version 9.2).

3. Results

Of the 853 study participants 51.8% were male, mean age at baseline was 48.1 ± 6.8 years, 34.7% originated from Europe/America, 26.4% from the Middle East, 22.0% from Yemen, and 17.0% were from North Africa, In total, 40.2% were ever-smokers, with a mean systolic blood pressure of 126.0 ± 18.3 mmHg and mean diastolic blood pressure of 82.9 ± 10.9 mmHg. By the end of the 24-year follow-up cumulative diabetes incidence was 31.5% and that of pre-diabetes was 33.2%.

Table 1 presents the baseline characteristics stratified based on glycemic categories. Individuals in Group-A were the youngest and had the lowest mean BMI for both men and women. The groups differed in sex distribution, blood pressure values, mean glucose, fasting insulin levels, HOMA-IR levels and triglycerides. No differences were noted between the study groups in ethnic origin distribution, smoking habits, mean total cholesterol, and mean HDL-cholesterol in both men and women.

Table 2 presents the progression of the cohort to prediabetes and diabetes between baseline and follow-up by OGTT 1 and 2-h study groups. While progression to diabetes was 19.6% in Group-A [1-h level ⩽155 mg/dL (8.6 mmol/l) and 2-h below140 mg/dL (7.8 mmol/l)], it reached 62.2% in Group D [1-h level >155 mg/dL (8.6 mmol/l) and 2-h 140–199 mg/dL (7.8–11.0 mmol/l)] with 50% of individuals having progressed to diabetes in the two intermediate Groups (B and C). The percent progressing to prediabetes (IFG, IGT or IFG + IGT) was 34.1, 33.5, 29.2 and 27.8, for individuals belonging to Groups A, B, C and D, respectively.

Table 3 presents the multinomial logistic regression models for prediabetes and for diabetes cumulative 24-y risk, according to baseline glycemic categories. Since Group C included only 24 individuals, it was combined with Group D, which has the same 2-h-PG criteria of 7.8–11.0 mmol/l (140–199 mg/dl). Group B had a significantly elevated risk, compared to Group A, for both diabetes and prediabetes outcomes (OR: 4.35, 95%CI 2.50–7.73, and OR: 1.87, 95%CI 1.09–3.26,

Table 1 – Baseline characteristics of the study sample by OGTT 1 and 2-h group (*N* = 853 with F-U diabetes incidence data).

	Group-A 1 h ⩽155 mg/dl and 2 h <140 mg/dl	Group-B 1 h >155 mg/dl and 2 h <140 mg/dl	Group-C 1 h ⩽155 mg/dl and 2 h = 140–199 mg/dl	Group-D 1 h >155 mg/dl and 2 h = 140–199 mg/dl	p
N (raw percent)	551 (64.6)	188 (22.0)	24 (2.8)	90 (10.6)	
Sex (%)					<0.001
Men	43.9	71.3	62.5	56.7	
Women	56.1	28.7	37.5	43.3	
Age (years)	47.4 ± 6.6	49.1 ± 6.6	50.5 ± 8.3	50.0 ± 7.0	<0.001
Ethnic origin (%)					0.19
Europe/America	34.5	30.3	41.7	43.3	
Middle East	27.6	22.9	29.2	25.6	
North Africa	17.4	17.0	12.5	14.4	
Yemen	20.5	30.3	16.7	16.7	
BMI (kg/m^2)					
Men	25.1 ± 3.2	25.6 ± 3.1	26.5 ± 3.8	27.8 ± 3.7	<0.001
Women	25.3 ± 3.4	27.2 ± 5.5	26.8 ± 5.2	27.0 ± 4.7	<0.001
Blood pressure (mmHg)					
Systolic	123.7 ± 17.8	130.0 ± 18.4	128.3 ± 15.8	130.7 ± 20.1	<0.001
Diastolic	81.6 ± 10.8	85.2 ± 10.4	84.0 ± 10.4	85.9 ± 11.6	<0.001
Smoking (%)					0.58
Current or past	40.3	43.1	37.5	34.4	
Never	59.7	56.9	62.5	65.6	
Biochemistry (mean ± SD)					
Fasting glucose, mg/dL	94.6 ± 9.1	102.4 ± 9.4	95.4 ± 8.2	105.8 ± 9.5	<0.001
1-h glucose, mg/dL	114.0 ± 22.3	182.7 ± 22.6	137.7 ± 18.5	196.2 ± 26.9	<0.001
2-h glucose, mg/dL	99.2 ± 16.7	111.7 ± 15.5	155.0 ± 17.3	159.2 ± 15.4	<0.001
Insulin – fasting, U/mL*	14.6 ± 11.0	15.5 ± 8.8	15.4 ± 10.2	18.0 ± 9.8	0.1
Total cholesterol, mg/dl*	214.7 ± 52.7	220.2 ± 55.6	218.3 ± 44.7	218.0 ± 55.6	0.74
HDL-cholesterol, mg/dl*					
Men	41.1 ± 10.9	42.5 ± 11.1	45.7 ± 11.7	41.1 ± 14.0	0.41
Women	48.7 ± 12.1	47.1 ± 11.9	41.3 ± 11.1	48.8 ± 13.8	0.88
Triglycerides*, mg/dl (median) [IQR: 25%; 75%]	100 [70; 140]	115 [80; 160]	120 [100; 160]	128 [80; 175]	<0.007
HOMA-IR (median)*	2.96	3.37	3.36	4.21	<0.001

* N = 668; IQR = interquartile range.

Table 2 – Progression in glucose metabolic state of 853 individuals free of diabetes in phase-II with follow up data in Phase-III.

Phase II			Phase III				
Fasting plasma glucose	1-h plasma glucose	2-h plasma glucose	Normoglycemic	IFG	IGT	IGT + IFG	Diabetes
<126 mg/dl	⩽155 mg/dl	A: <140 mg/dl	255 (46.3)	125 (22.7)	30 (5.4)	33 (6.0)	108 (19.6)
		C: 140–199 mg/dl	5 (20.8)	1 (4.2)	4 (16.7)	2 (8.3)	12 (50.0)
	>155 mg/dl	B: <140 mg/dl	32 (17.0)	35 (18.6)	15 (8.0)	13 (6.9)	93 (49.5)
		D: 140–199 mg/dl	9 (10.0)	16 (17.8)	3 (3.3)	6 (6.7)	56 (62.2)
Total:			301 (35.3)	177(20.8)	52 (6.1)	54 (6.3)	269 (31.5)

respectively), adjusted for sex and age, smoking, BMI, blood pressure, fasting blood glucose and insulin. This emphasizes the importance of the 1-h-PG >8.6 mmol/l (155 mg/dl) as a prognostic biomarker when the 2-h-BG is still within the normal range less than 7.8 mmol/l (140 mg/dl). Groups C + D with the 2-h-PG 7.8–11.0 mmol/l (140–199 mg/dl) showed an expected increased risk for progression to both outcomes when compared to Group A.

Fig. 2a and b show that the power of the1-h PG is greater than the 2-h PG in predicting cumulative incident diabetes (AUC-1-h PG: 0.736 [95%CI: 0.699–0.773] vs. AUC-2-h: 0.707 [95%CI: 0.669–0.745]) and prediabetes (AUC1-h PG: 0.659 [95% CI: 0.615–0.703] vs. AUC2-h: 0.632, [95%CI: 0.587–0.677]).

4. Discussion

We have found in this long-term follow-up study that individuals with a 1 h glucose level >8.6 mmol/l (155 mg/dl) are at significantly increased risk for progression to prediabetes and diabetes. Furthermore, the risk for progression to both conditions is increased when the 2-h level achieves glucose levels in the IGT range. Prediabetes is associated with progression to diabetes and risk for cardiovascular disease as well as other complications [18,19].

Dysglycemic conditions occur as a continuous process [6]. Therefore, defining the latter by absolute diagnostic criteria may inadvertently limit earlier intervention when this may be particularly effective [20]. Determining the most suitable approach for identifying those at risk for progression to diabetes has received considerable interest with studies suggesting that an elevated 30 or 60 min post-load glucose value during the OGTT could be more effective than a 2 h level >7.8 mmol/l (140 mg/dl) [5,6,10,21–24]. In a publication of the Israeli GOH study we previously showed that the fasting and the 1-h OGTT glucose levels are as predictive for 20-year cumulative diabetes incidence as the 2-h OGTT glucose [25]. In the San Antonio Metabolism and RISC Trials [26,27], β-cell function were impaired in those with normal glucose tolerance (NGT) having the highest 2-h OGTT glucose values. Risk could be further stratified with the 1-h OGTT threshold of 155 mg/dl (8.6 mmol/l). Furthermore, the Botnia Study and Malmö Prevention Project (MPP) demonstrated that an elevated 1-h-plasma glucose predicted progression to T2DM [10] leading the investigators to recommend that the latter could identify high-risk individuals for intervention to reduce the progression to T2DM.

A 1-h glucose ⩾155 mg/dl (8.6 mmol/l) in NGT individuals has been associated with a worse cardiovascular risk profile and subclinical organ damage [11,28–30] independent of other predictors for cardiovascular events. A significant correlation was demonstrated between an elevated 1-h post-load plasma glucose and left ventricular mass (LVM) in hypertensive patients [29]. Men with NGT having a 1-h glucose ⩾155 mg/dl (8.6 mmol/l) had a significantly higher prevalence of left ventricular hypertrophy (LVH) similar to those with IGT and type 2 diabetes [29]. These observations could underlie our recent finding that a 1-h glucose value >8.6 mmol/l (155 mg/dl) predicted mortality even when the 2-h level was <7.8 mmol/l (140 mg/dl) [7]. Of importance, mortality was

Table 3 – Multinomial logistic regression models for prediabetes and diabetes 24-year cumulative incidence according to baseline glycemic group.

Glycemic group	Diabetes			Prediabetes		
	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
Glucose in mg/dl	OR (95%CI)	OR (95%CI)	OR (95%CI)	OR (95%CI)	OR (95%CI)	OR (95%CI)
A: 1 h ⩽155 and 2 h <140	1.00	1.00	1.00	1.00	1.00	1.00
B: 1 h >155 and 2 h <140	6.77 (4.26–11.00)	4.79 (2.92–7.99)	4.35 (2.50–7.73)	2.42 (1.51–3.92)	2.09 (1.28–3.45)	1.87 (1.09–3.26)
C + D: 2 h = 140–199	11.62 (6.39–22.49)	7.38 (3.92–14.72)	5.64 (2.74–12.33)	2.86 (1.50–5.70)	2.39 (1.23–4.85)	1.73 (0.82–3.82)

Adjusted for:
Model 1: Sex and age (N = 853).
Model 2: Sex, age, smoking, BMI, systolic, diastolic blood pressure, and fasting BG (N = 843).
Model 3: Like model 2 adjusted for fasting insulin (N = 656).

shown to rise significantly independent of the 1-h value with the occurrence of IGT. The results of the present study, in conjunction with the cardiovascular surrogate marker findings described above, may explain the increased mortality in those with a 1-h level >8.6 mmol/l (155 mg/dl) which also worsened when IGT developed [7].

Insulin secretion retained with younger age is vital in prediabetes with enhanced insulin secretion predicting regression to normal glucose regulation in the US Diabetes Prevention Program [31]. Prediabetes represents "a high-risk state" for diabetes especially in those who remained with prediabetes despite intensive lifestyle intervention during the Diabetes Prevention Program Observation Study. Even transient reversion to normal glucose regulation was associated with a significantly reduced risk of future diabetes [32]. Finally, as the long-term Diabetes Prevention Program Observation Study demonstrated a decline [33] in benefit years after the initial intervention, earlier lifestyle intervention during the so-called "pre-prediabetic" phase, prior to the development of prediabetes, could prove even more durable and provide greater reduction in morbidity and mortality. It is therefore conceivable that the effective intervention demonstrated by the U.S. Diabetes Prevention Program [34] in those with IGT could be further improved and sustained with long-term follow-up by identifying those at risk much earlier in the trajectory to diabetes in the lengthy "pre-prediabetic" phase when β-cells are yet more functional.

Some limitations and strengths of the current study need to be acknowledged. OGTT was performed once at baseline and glucose and insulin values may have been subjected to the regression dilution bias [35]. In addition, no temporal relationship between abnormal 1-h and abnormal 2-h glucose levels could be studied. Although no longer utilized, the 100 gram glucose load was used routinely when this study was initiated during the 1980's with consistent results over a 33 year period [7]. Furthermore, as discussed above, because the additional glucose load had little effect on the glucose level [15] but enhanced the insulin response [16], it was not expected to influence the use of the clinical threshold of 200 mg/dl for diabetes.

Among the advantages of this study are the nature of the randomly selected cohort, the length of the follow- up, and the high reliability of the parameters – all measured uniformly using one laboratory.

In summary, given the growing body of literature, including observations from the current study, identifying individuals at risk with a 1-h post-load glucose level during an OGTT is recommended. As the risk for diabetes and mortality associated with a 1-h level >8.6 mmol/l (155 mg/dl) is increased and is even greater in the presence of IGT, high-risk individuals with an elevated 1-h level should be referred for lifestyle intervention preferably before IGT occurs. In view of the present findings, consistent with results from a variety of studies, evaluating the benefit of earlier intervention in those with an elevated 1-h level >8.6 mmol/l (155 mg/dl) with lifestyle intervention is warranted. Furthermore, the 1-h OGTT could potentially have public health implications as it would likely become more acceptable in general practice and thereby increase the identification of those at heightened risk

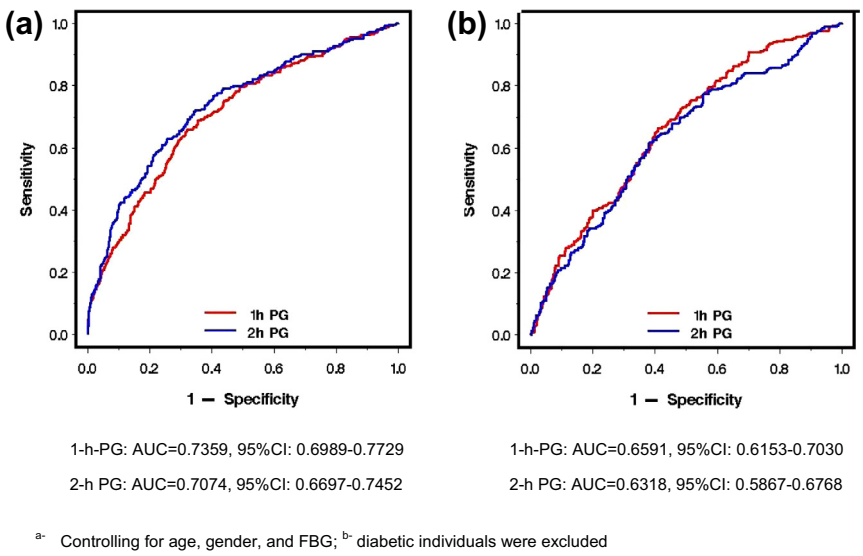

Fig. 2 – AUC of 1-h (red) and 2-h (blue) PG. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

for progression to diabetes, cardiovascular disease and mortality.

Contributions

Michael Bergman: conception of the study, interpretation of the results, drafting of the paper.

Angela Chetrit: Data collection, statistical analysis, drafting of the paper.

Jesse Roth: Data collection, interpretation of the results, drafting of the paper.

Ram Jagannathan: interpretation of the results, drafting of the paper.

Mary Sevick: interpretation of the results, drafting of the paper.

Rachel Dankner: Conception of the study, data collection, interpretation of the results, drafting of the paper.

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Conflict of interest

None to declare.

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